Tutorial Questions （Week 7）

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Outline

- Basic knowledge review
- Basic Problems with Answers 4.8, 4.9
- Basic Problems 4.23
- Advanced Problems 4.39, 4.40
- Q&A

Basic knowledge review

- Fourier transform formula

- Properties of Fourier transforms (integral, linear, parity, dual)

Fourier transform formula

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$$

Inverse Fourier transform equation
(synthesis equation)

$$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$$

Fourier transform equation

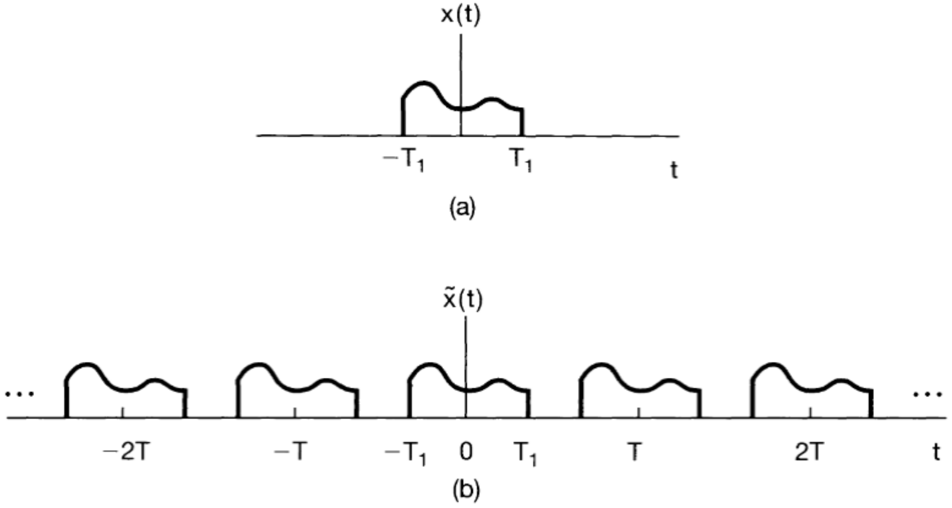


Figure 4.3 (a) Aperiodic signal $x(t)$; (b) periodic signal $\tilde{x}(t)$, constructed to be equal to $x(t)$ over one period.

Continuous-Time Fourier Transform (CTFT)

Section	Property	Aperiodic signal	Fourier transform
		$x(t)$	$X(j\omega)$
		$y(t)$	$Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0}X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t}x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi}\int_{-\infty}^{+\infty}X(j\theta)Y(j(\omega-\theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^{t} x(t)dt$	$\frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j\frac{d}{d\omega}X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \Re e\{X(j\omega)\} = \Re e\{X(-j\omega)\} \\ \Im m\{X(j\omega)\} = -\Im m\{X(-j\omega)\} \\ \lvert X(j\omega)\rvert = \lvert X(-j\omega)\rvert \\ \sphericalangle X(j\omega) = -\sphericalangle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}v\{x(t)\}$ [$x(t)$ real] $x_o(t) = \mathcal{O}d\{x(t)\}$ [$x(t)$ real]	$\Re e\{X(j\omega)\}$ $j\Im m\{X(j\omega)\}$

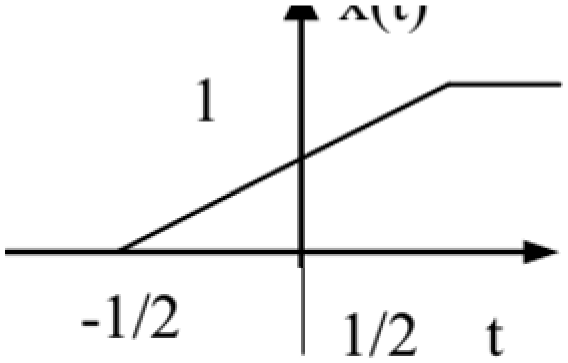
4.3.7 Parseval's Relation for Aperiodic Signals

$$\int_{-\infty}^{+\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{+\infty} |X(j\omega)|^2 d\omega$$

Signal	Fourier transform	Fourier series coefficients (if periodic)
$\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$2\pi \sum_{k=-\infty}^{+\infty} a_k \delta(\omega - k\omega_0)$	a_k
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$	$a_1 = 1$ $a_k = 0$, otherwise
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	$a_1 = a_{-1} = \frac{1}{2}$ $a_k = 0$, otherwise
$\sin \omega_0 t$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	$a_1 = -a_{-1} = \frac{1}{2j}$ $a_k = 0$, otherwise
$x(t) = 1$	$2\pi\,\delta(\omega)$	$a_0 = 1, \quad a_k = 0, \ k \neq 0$ (this is the Fourier series representation for any choice of $T > 0$)
Periodic square wave $x(t) = \begin{cases} 1, & \lvert t\rvert < T_1 \\ 0, & T_1 < \lvert t\rvert \leq \frac{T}{2} \end{cases}$ and $x(t + T) = x(t)$	$\sum_{k=-\infty}^{+\infty} \frac{2 \sin k\omega_0 T_1}{k} \delta(\omega - k\omega_0)$	$\frac{\omega_0 T_1}{\pi} \operatorname{sinc}\left(\frac{k\omega_0 T_1}{\pi}\right) = \frac{\sin k\omega_0 T_1}{k\pi}$
$\sum_{n=-\infty}^{+\infty} \delta(t - nT)$	$\frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$	$a_k = \frac{1}{T}$ for all k
$x(t) \begin{cases} 1, & \lvert t\rvert < T_1 \\ 0, & \lvert t\rvert > T_1 \end{cases}$	$\frac{2 \sin \omega T_1}{\omega}$	—
$\frac{\sin Wt}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \lvert\omega\rvert < W \\ 0, & \lvert\omega\rvert > W \end{cases}$	—
$\delta(t)$	1	—
$u(t)$	$\frac{1}{j\omega} + \pi\,\delta(\omega)$	—
$\delta(t - t_0)$	$e^{-j\omega t_0}$	—
$e^{-at}u(t), \Re e\{a\} > 0$	$\frac{1}{a + j\omega}$	—
$te^{-at}u(t), \Re e\{a\} > 0$	$\frac{1}{(a + j\omega)^2}$	—
$\frac{t^{n-1}}{(n-1)!} e^{-at}u(t)$, $\Re e\{a\} > 0$	$\frac{1}{(a + j\omega)^n}$	—

4.8. Consider the signal

$$x(t) = \begin{cases} 0, & t < -\frac{1}{2} \\ t + \frac{1}{2}, & -\frac{1}{2} \leq t \leq \frac{1}{2}. \\ 1, & t > \frac{1}{2} \end{cases}$$



(a) Use the differentiation and integration properties in Table 4.1 and the Fourier transform pair for the rectangular pulse in Table 4.2 to find a closed-form expression for $X(j\omega)$.

(b) What is the Fourier transform of $g(t) = x(t) - \frac{1}{2}$?

5) Differentiation/Integration

$$x(t)\begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases} \qquad \frac{2\sin\omega T_1}{\omega}$$

$$\frac{dx(t)}{dt} \stackrel{F}{\longleftrightarrow} j\omega X(j\omega)$$

$$\int_{-\infty}^{t} x(\tau)d\tau \stackrel{F}{\longleftrightarrow} \frac{1}{j\omega}X(j\omega) + \pi \underset{\substack{\uparrow \\ \text{DC term}}}{X(j0)}\delta(\omega)$$

4.8. We may express this signal as

$$x(t) = \int_{-\infty}^{t} y(t)dt$$

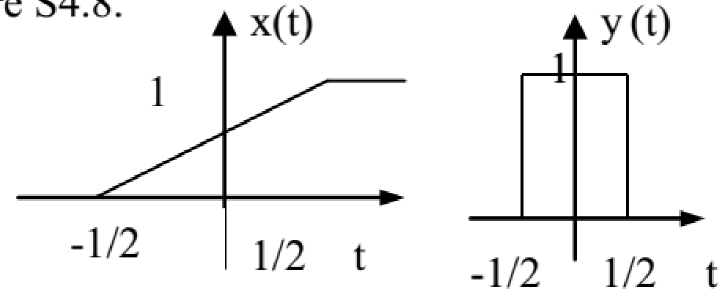
where $y(t)$ is the rectangular pulse shown in Figure S4.8. Using the integration property of the Fourier transform, we have

$$x(t) \stackrel{FT}{\longleftrightarrow} X(j\omega) = \frac{1}{j\omega}Y(j\omega) + \pi Y(j0)\delta(\omega)$$

We know from Table 4.2 that

$$Y(j\omega) = \frac{2\sin(\omega/2)}{\omega}$$

Figure S4.8.



Therefore,

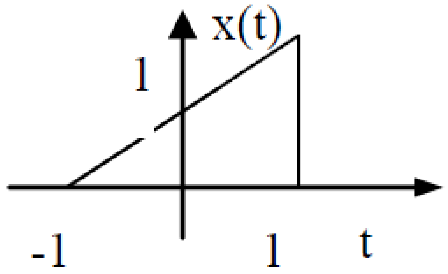
$$X(j\omega) = \frac{2\sin(\omega/2)}{j\omega^2} + \pi\delta(\omega)$$

(b) If $g(t) = x(t) - 1/2$, then the Fourier transform $G(j\omega)$ of $g(t)$ is given by

$$G(j\omega) = X(j\omega) - (1/2)2\pi\delta(\omega) = \frac{2\sin(\omega/2)}{j\omega^2}$$

4.9. Consider the signal

$$x(t) = \begin{cases} 0, & |t| > 1 \\ (t+1)/2, & -1 \le t \le 1 \end{cases}.$$



(a) With the help of Tables 4.1 and 4.2, determine the closed-form expression for $X(j\omega)$.

(b) Take the real part of your answer to part (a), and verify that it is the Fourier transform of the even part of $x(t)$. $\mathcal{E}v\{x(t)\}$=(x(t)+x(-t))/2

(c) What is the Fourier transform of the odd part of $x(t)$?

$\boxed{\text{5) Differentiation/Integration}}$

$$\frac{dx(t)}{dt} \stackrel{F}{\longleftrightarrow} j\omega X(j\omega)$$

$$\int_{-\infty}^{t} x(\tau)\,d\tau \stackrel{F}{\longleftrightarrow} \frac{1}{j\omega} X(j\omega) + \pi X(j0)\delta(\omega)$$

DC term

$$x(t)\begin{cases} 1, & |t| < T_1 \\ 0, & |t| > T_1 \end{cases} \qquad \frac{2\sin\omega T_1}{\omega}$$

$$u(t) \qquad \frac{1}{j\omega} + \pi\delta(\omega)$$

Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}v\{x(t)\}$	$[x(t)$ real$]$	$\mathcal{R}e\{X(j\omega)\}$
	$x_o(t) = \mathcal{O}d\{x(t)\}$	$[x(t)$ real$]$	$j\mathcal{I}m\{X(j\omega)\}$

4.9. (a) The signal $x(t)$ is plotted in Figure S4.9.

We see that this signal is very similar to the one considered in the previous problem. In fact we may again express the signal $x(t)$ in terms of the rectangular pulse $y(t)$ shown above as follows

$$x(t) = \int_{-\infty}^{t} y(t)dt - u\left(t - \frac{1}{2}\right).$$

Using the result obtained in part (a) of the previous problem, the Fourier transform $X(j\omega)$ of $x(t)$ is

$$X(j\omega) = \frac{2\sin(\omega/2)}{j\omega^2} + \pi\delta(\omega) - \mathcal{FT}\left\{u\left(t - \frac{1}{2}\right)\right\} = \frac{\sin\omega}{j\omega^2} - \frac{e^{-j\omega}}{j\omega}$$

(b) The even part of $x(t)$ is given by

$$\mathcal{E}v\{x(t)\} = \frac{x(t) + x(-t)}{2}.$$

This is as shown in the Figure S4.9. Therefore,

$$\mathcal{FT}\{\mathcal{E}v\{x(t)\}\} = \frac{\sin\omega}{\omega}.$$

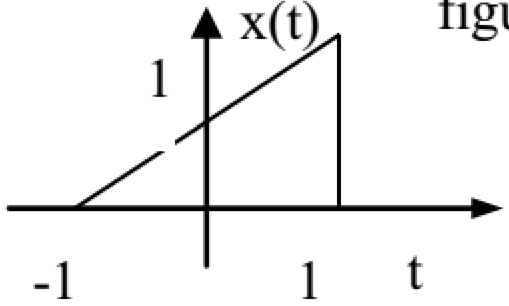
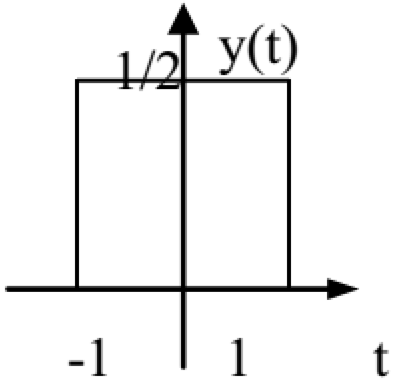


figure S4.9



Now the real part of the answer to part (a) is

$$Re\left\{-\frac{e^{-j\omega}}{j\omega}\right\} = (1/\omega)Re\left\{j(\cos\omega - j\sin\omega)\right\} = \frac{\sin\omega}{\omega}$$

(c) The Fourier transform of the odd part of $x(t)$ is same as j times imaginary part of the answer to part (a). We have

$$Im\left\{\frac{\sin\omega}{j\omega^2} - \frac{e^{-j\omega}}{j\omega}\right\} = -\frac{\sin\omega}{\omega^2} + \frac{\cos\omega}{\omega}$$

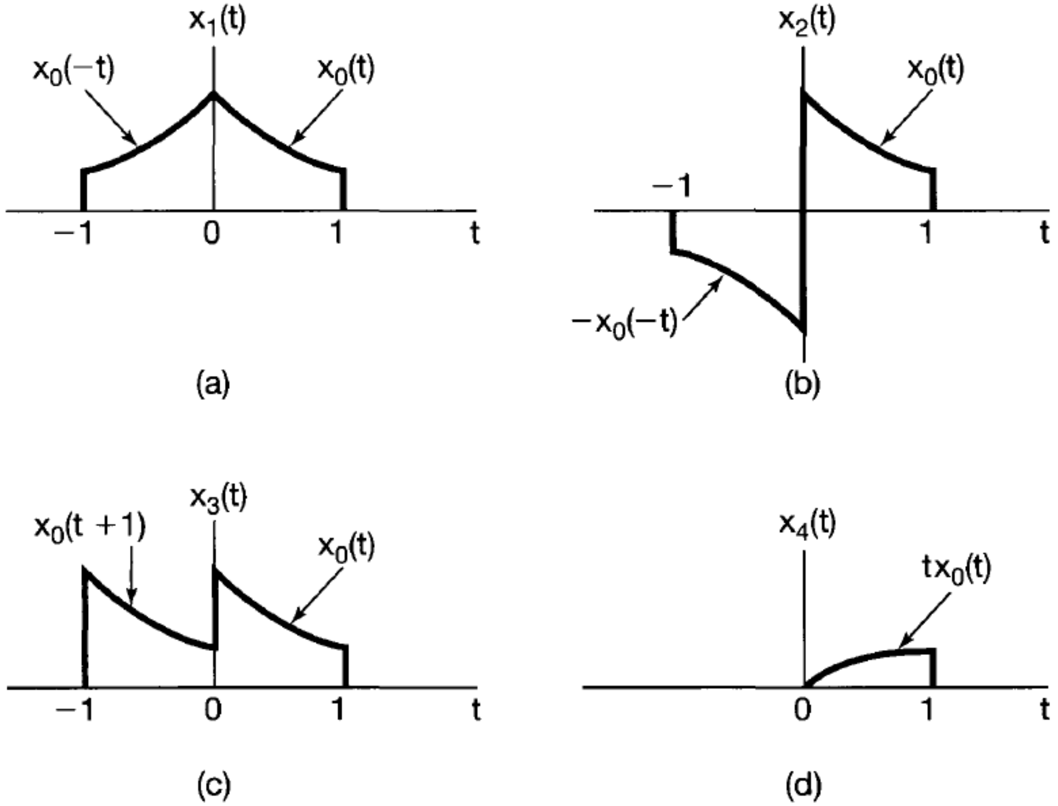
4.23. Consider the signal

$e^{-at}u(t), \Re e\{a\} > 0 \qquad \frac{1}{a + j\omega}$

$$x_0(t) = \begin{cases} e^{-t}, & 0 \le t \le 1 \\ 0, & \text{elsewhere} \end{cases}.$$

$$X_0(j\omega) = \frac{1-e^{-(1+j\omega)}}{1+j\omega}$$

Determine the Fourier transform of each of the signals shown in Figure P4.23. You should be able to do this by explicitly evaluating *only* the transform of $x_0(t)$ and then using properties of the Fourier transform.



Time Reversal	$x(-t)$	$X(-j\omega)$
Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0}X(j\omega)$
Differentiation in Frequency	$tx(t)$	$j\frac{d}{d\omega}X(j\omega)$

Figure P4.23

4.23 For the given signal $x_0(t)$, we use the Fourier transform analysis eq. (4.8) to evaluate the corresponding Fourier transform

$$X_0(j\omega) = \frac{1 - e^{-(1+j\omega)}}{1 + j\omega}$$

(i) We know that

$$x_1(t) = x_0(t) + x_0(-t)$$

Using the linearity and time reversal properties of the Fourier transform we have

$$X_1(j\omega) = X_0(j\omega) + X_0(-j\omega) = \frac{2 - 2e^{-1}\cos\omega - 2\omega e^{-1}\sin\omega}{1 + \omega^2}$$

(ii) We know that

$$x_2(t) = x_0(t) - x_0(-t)$$

Using the linearity and time reversal properties of the Fourier transform we have

$$X_2(j\omega) = X_0(j\omega) - X_0(-j\omega) = \frac{-2\omega + 2e^{-1}\sin\omega + 2\omega e^{-1}\cos\omega}{1 + \omega^2}$$

(iii) We know that

$$x_3(t) = x_0(t) + x_0(t+1)$$

Using the linearity and time shifting properties of the Fourier transform we have

$$X_3(j\omega) = X_0(j\omega) + e^{j\omega} X_0(-j\omega) = \frac{1 + e^{j\omega} - e^{-1}(1 + e^{-j\omega})}{1 + j\omega}$$

(iv) We know that

$$x_4(t) = t x_0(t)$$

Using the differentiation in frequency property

$$X_4(j\omega) = j \frac{d}{d\omega} X_0(j\omega)$$

Therefore

$$X_4(j\omega) = \frac{1 - 2e^{-1}e^{-j\omega} - j\omega e^{-1}e^{-j\omega}}{(1 + j\omega)^2}$$

4.39. Suppose that a signal $x(t)$ has Fourier transform $X(j\omega)$. Now consider another signal $g(t)$ whose shape is the same as the shape of $X(j\omega)$; that is,

$$g(t) = X(jt).$$

(a) Show that the Fourier transform $G(j\omega)$ of $g(t)$ has the same shape as $2\pi x(-t)$; that is, show that

$$G(j\omega) = 2\pi x(-\omega).$$

(b) Using the fact that

$$\mathfrak{F}\{\delta(t+B)\} = e^{jB\omega}$$

in conjunction with the result from part (a), show that

$$\mathfrak{F}\{e^{jBt}\} = 2\pi\,\delta(\omega - B).$$

4.39 From the Fourier analysis equation. We have

$$G(j\omega) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} X(jt) e^{-j\omega t} dt \quad (S4.39 - 1)$$

Also for the Fourier Transform equation, we have

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

Switching the variables t and ω we have

$$x(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(jt) e^{j\omega t} dt$$

We may also write this equation as

$$2\pi x(-\omega) = \int_{-\infty}^{\infty} X(jt) e^{-j\omega t} dt$$

Substituting this equation in eq.(S4.39-1), we obtain

$$G(j\omega) = 2\pi x(-\omega)$$

4.40. Use properties of the Fourier transform to show by induction that the Fourier transform of

$$x(t) = \frac{t^{n-1}}{(n-1)!} e^{-at} u(t), \quad a > 0,$$

is

$$\frac{1}{(a + j\omega)^n}.$$

$e^{-at}u(t), \mathcal{R}e\{a\} > 0$	$\dfrac{1}{a + j\omega}$	Mathematical induction
$te^{-at}u(t), \mathcal{R}e\{a\} > 0$	$\dfrac{1}{(a + j\omega)^2}$	
Differentiation in Frequency	$tx(t)$	$j\dfrac{d}{d\omega}X(j\omega)$

Mathematical Induction

Mathematical induction (or weak mathematical induction) is a method to prove or establish mathematical statements, propositions, theorems, or formulas for all natural numbers 'n $\geq$1.'

Principle

It involves two steps:

1. **Base Step**: It proves whether a statement is true for the initial value (n), usually the smallest natural number in consideration.

2. **Inductive Step**: It proves that when the statement is true for the natural number (n) (or n^{th} iteration), it must also be true for the next number (n + 1) or $(n+1)^{th}$ iteration.

[Mathematical Induction - Principle with Steps, Proof, & Examples]

4.40
When n = 1, $x_1(t) = e^{-at}u(t)$ and $X_1(j\omega) = 1/(a + j\omega)$
When n = 2, $x_2(t) = te^{-at}u(t)$ and $X_2(j\omega) = 1/(a + j\omega)^2$
Now, let us assume that the given statement is true when n = m, that is,

$$x_m(t) = \frac{t^{m-1}}{(m-1)!}e^{-at}u(t) \overset{FS}{\longleftrightarrow} X_m(j\omega) = \frac{1}{(a+j\omega)^m}$$

For n = m + 1 we may use the differentiation in frequency property to write,

$$x_{m+1}(t) = \frac{t}{m}x_m(t) \overset{FS}{\longleftrightarrow} X_{m+1}(j\omega) = \frac{1}{m}j\frac{dX_m(j\omega)}{d\omega} = \frac{1}{(a+j\omega)^{m+1}}$$

This shows that if we assume that the given statement is true for n = m, then it is true for n = m + 1. Since we also shown that the given statement is true for n = 2, we may argue that is is ture for n = 2 + 1 = 3, n = 3 + 1 = 4 and so on. Therefore, the given statement is true for any n.

Thanks for Your Attendance

Q&A

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